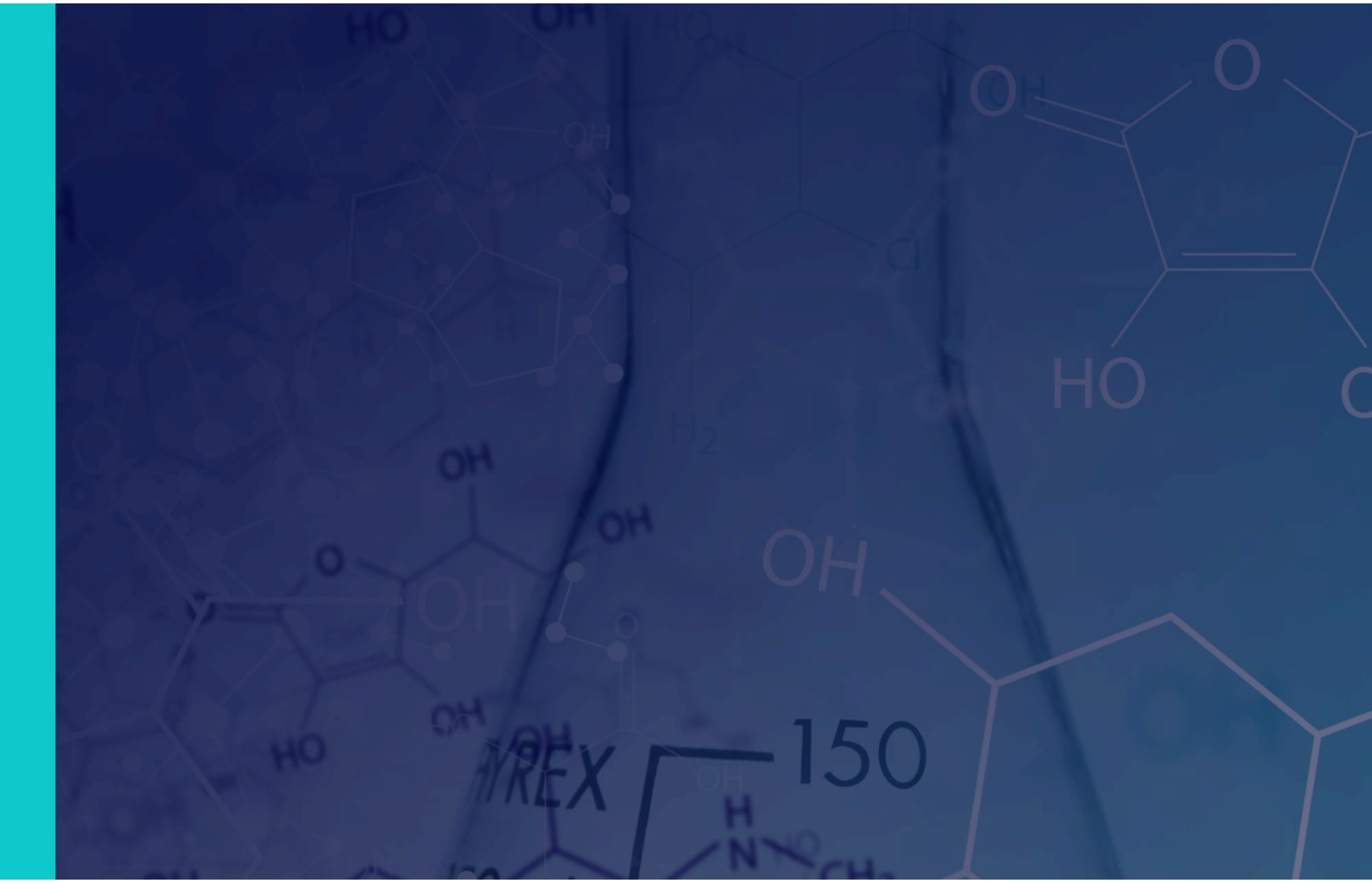


Canada High School

Grade 12 Chemistry

Summer 2024, Chapter 2 Notes

Table of Contents

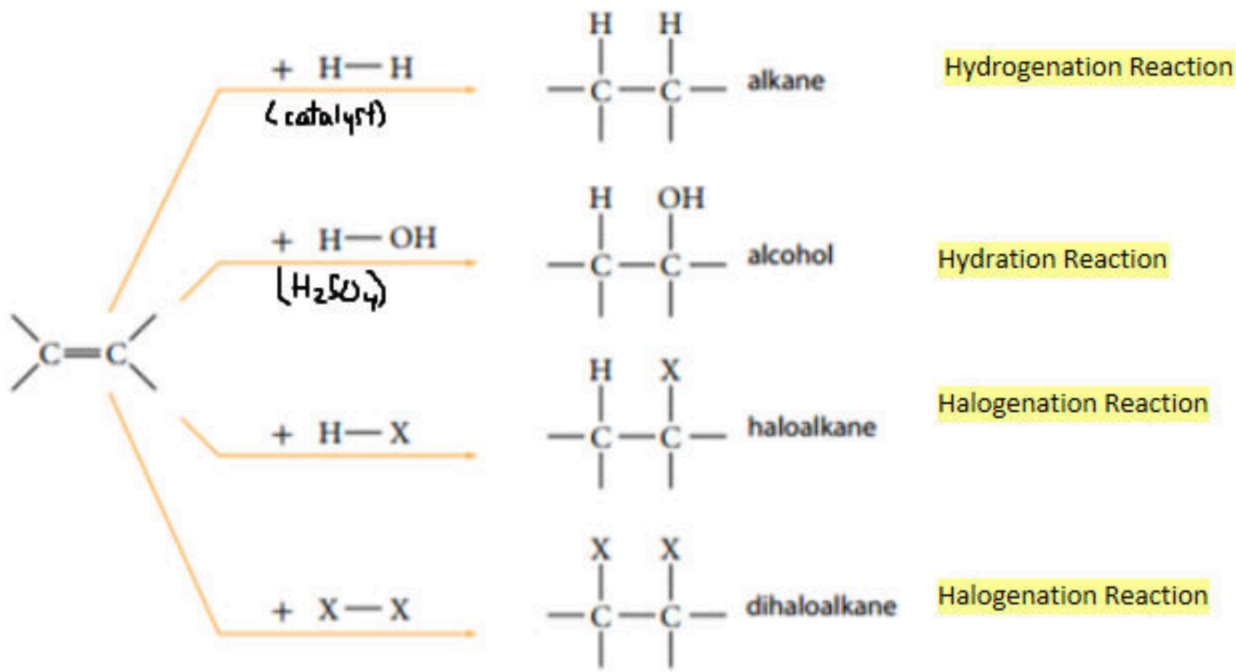
2. Organic Chemistry (Part 2: Chemical Reactions)

2.1 Addition Reactions

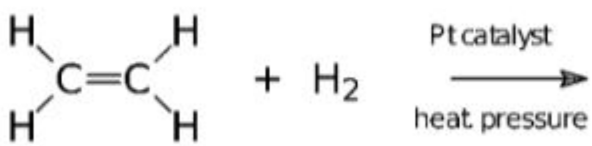
2.1.1

Addition Reactions

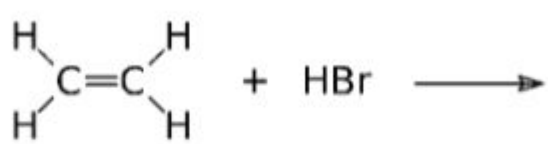
- an addition reaction is when atoms are being added across a carbon-carbon double or triple bond



Example: What is the product? (Draw the structural formula and provide its name)



Example: What is the product? (Draw the structural formula and provide its name)



2.1.2

Markovnikov's Rule

Predict the product of the following reaction:

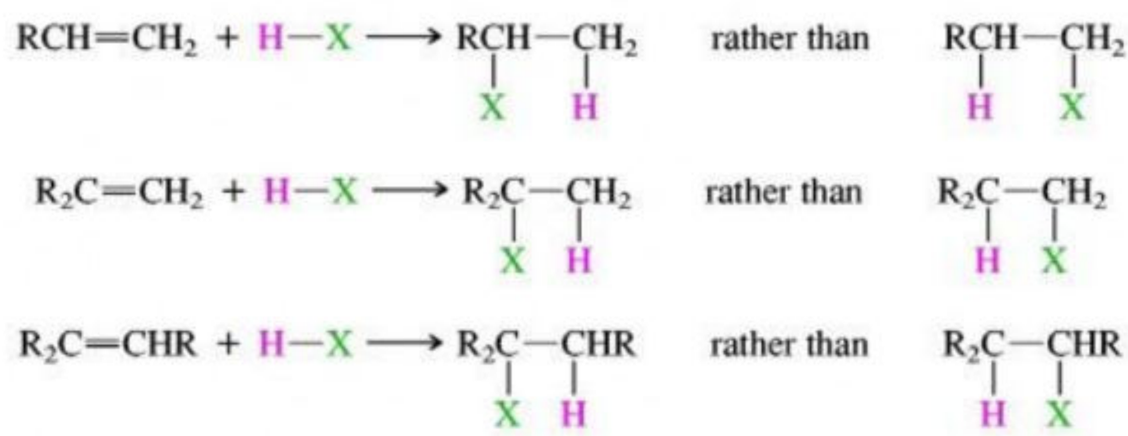
$$CH_3\text{-}CH{=}CH_2 \quad + \quad H_2O \xrightarrow{H_3O^+}$$

Markovnikov's rule: when more than one product is possible, the hydrogen will end up on the carbon that has more hydrogens bound to it.

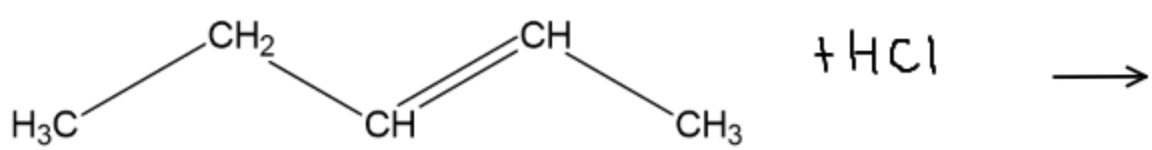
> **WIZE TIP**
> **A memory tip to help you remember that the carbon that already has more bonds to hydrogen will get another hydrogen is "the rich get richer".**

- Can get a major and minor product for the above reaction (the major one is the one that is formed most of the time where the hydrogen goes to the C that is bound to the most hydrogens)
- What is the major product of the above reaction?

When reacting with a hydrogen and an electronegative element, Markovnikov's rule applies:



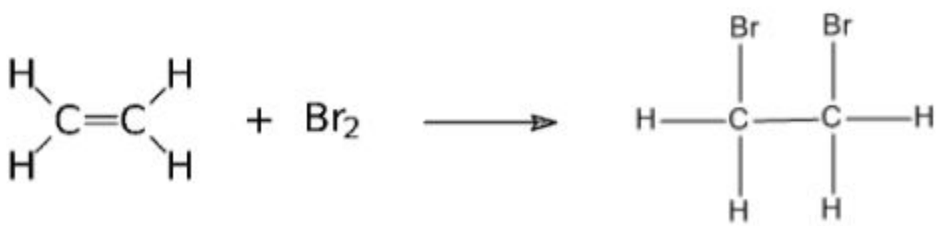
Example: Predict the product of the following reaction



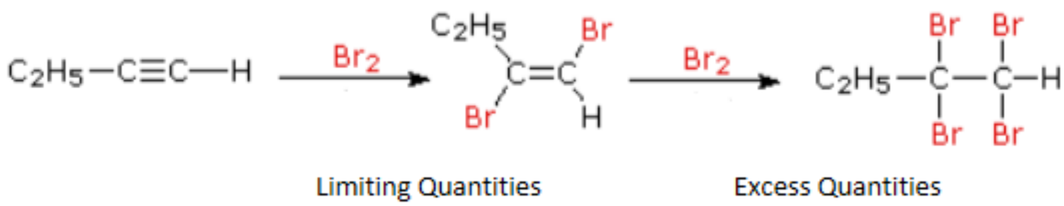
2.1.3

Addition Reactions For Alkenes Vs Alkynes

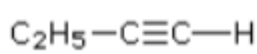
Alkenes + X_2 -> Dihaloalkanes



Alkynes Can React Differently Depending on if the X2 is in limiting or excess quantities:



Example: React the following molecule with HCl in limiting and excess quantities of HCl:



2.1.4

Draw out and name the product (indicate major and minor products if applicable).

2-methylpent-2-ene + HBr

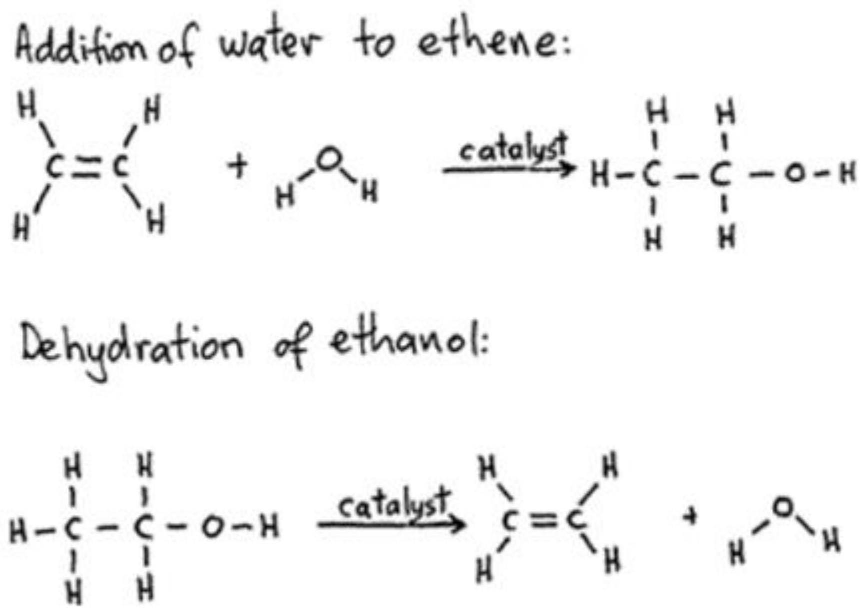
Enter the name of the major product below:

2.2 Elimination Reactions

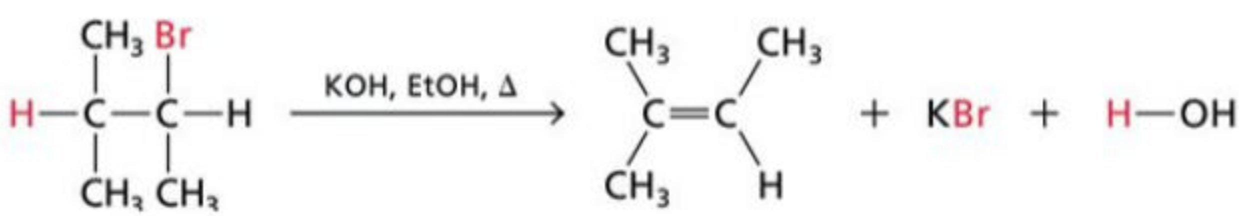
2.2.1

Elimination Reactions

- These reactions are when atoms are removed from a molecule to form a double bond (reverse of an addition reaction)



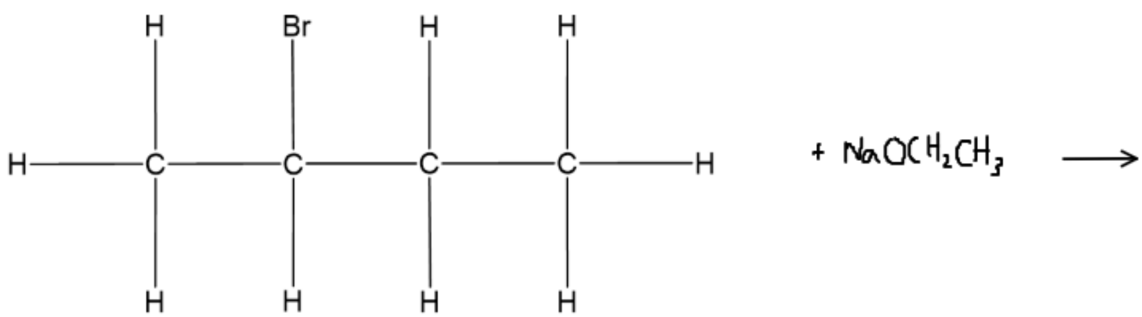
Another example of an elimination reaction:



2.2.2

Example: Elimination Reactions Forming More Than One Product

Predict the product in the following reaction:



WIZE TIP

The major product in an elimination reaction will be the one where the H is removed from the C atom with the most C-C bonds!

2.2.3

For the following reaction:

butan-2-ol $\xrightarrow{H_2SO_4}$

Draw out the reactant and product (and any major/minor products if applicable).

Name the major product in the answer box below:

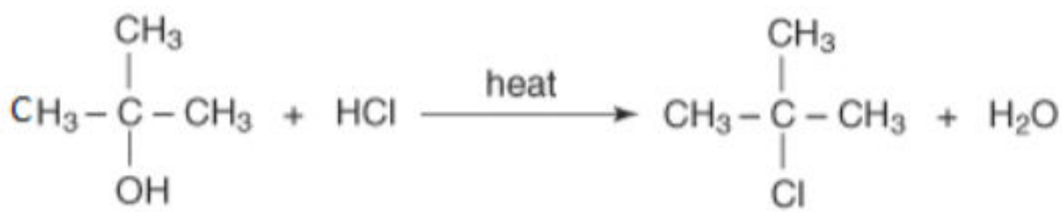
2.3 Substitution Reactions

2.3.1

Substitution Reactions

- This is a reaction where a hydrogen atom or functional group is replaced by a different atom or functional group

1) Alcohols can undergo substitution reactions when reacted with HX (where H represents an acidic proton, X=halogen)



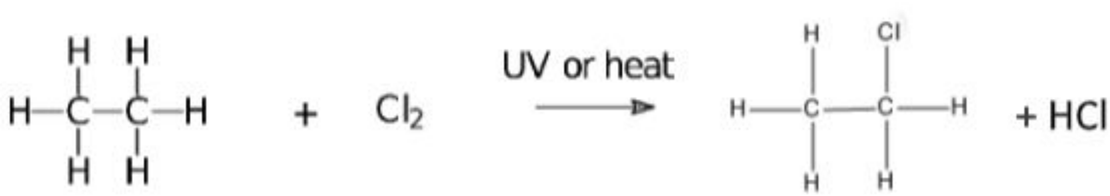
This reaction produces an alkyl halide.

2) An alkyl halide can undergo substitution reaction when reacted with OH^- (hydroxide ion, strong base)

$$CH_3\text{-}CH_2\text{-}CH_2\text{-}Cl + OH^- \rightarrow CH_3\text{-}CH_2\text{-}CH_2\text{-}OH + Cl^-$$

This reaction produces an __________.

3) Alkanes can undergo substitution with a halogen:



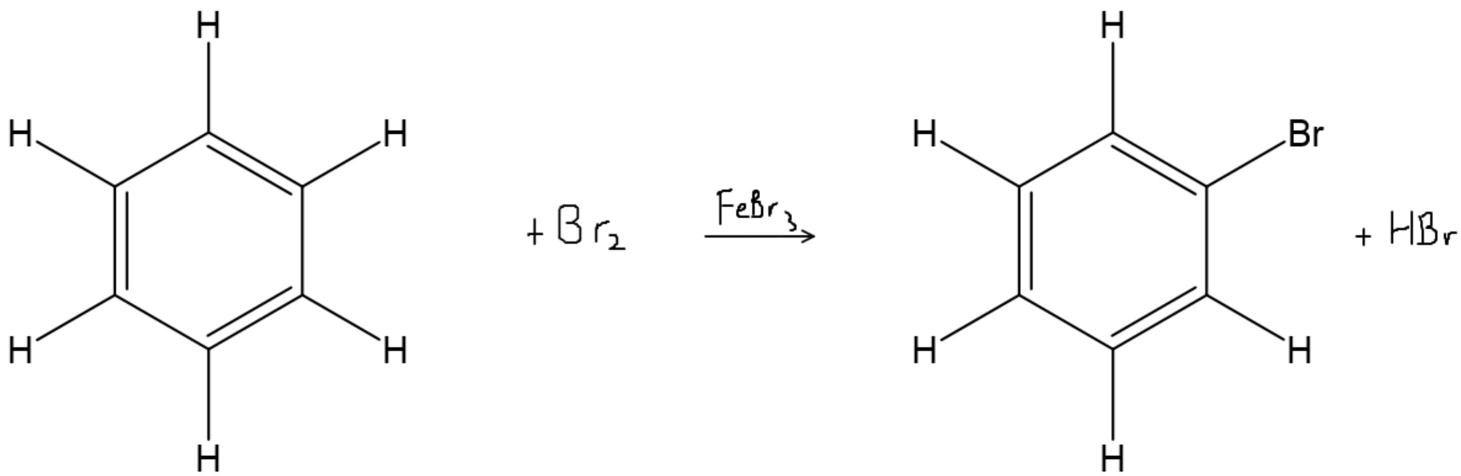
This reaction produces an alkyl halide.

This reaction can form multiple products (randomly) if the halogen is present in excess:

- It will result in a mixture of products

$$4CH_4 + 10Cl_2 \xrightarrow[\text{heat}]{\text{UV or}} CH_3Cl + CH_2Cl_2 + CHCl_3 + CCl_4 + 10HCl$$

4) Benzene can react with Cl_2 or Br_2 to form a halogenated benzene in a substitution reaction:



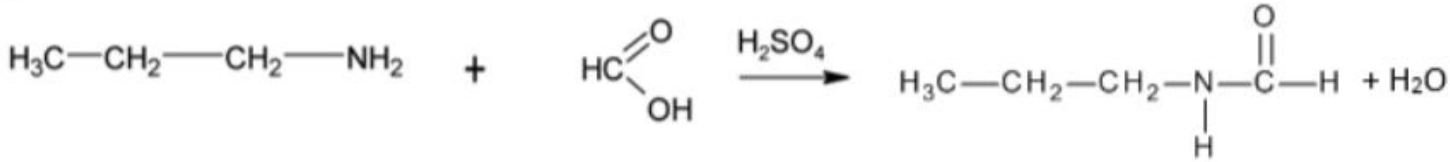
2.4 Condensation Reactions

2.4.1

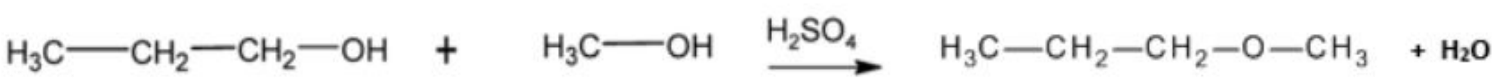
Condensation Reactions

- This is when two molecules combine to form one large molecule and a smaller molecule that's usually water

1) Amine + carboxylic acid react together to <u>form an amide</u> and water:

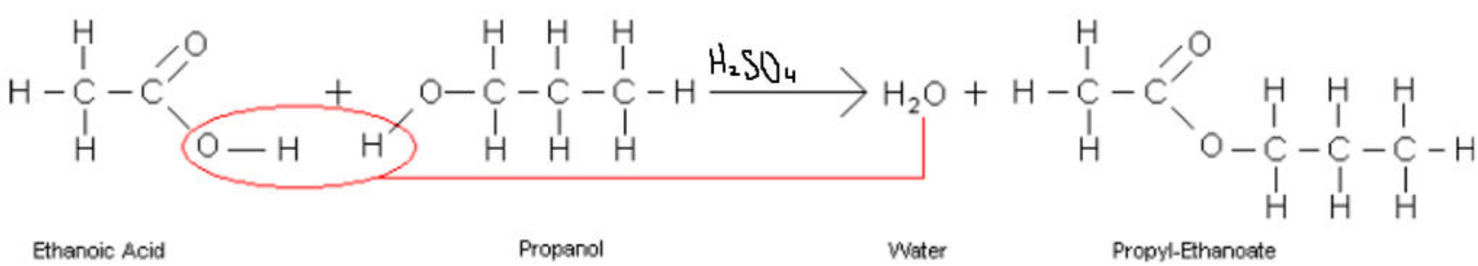


2) Two alcohols can react together to <u>form an ether</u> and water:



Review: Name each reactant and product.

3) A carboxylic acid + alcohol react together to <u>form an ester</u>:



- This reaction is also called **esterification** because we are forming an ester

2.4.2

Which of the features below is universal to ALL condensation reactions?

- An ester if formed
- An alcohol is a reactant
- Water is formed
- An amide is a product
- An Amine is a reactant

2.4.3

For the following reaction, draw out the reactants and product(s).

butanoic acid + ethanol $\xrightarrow{H_2SO_4}$

Provide the name of the (major) product in the answer box below:

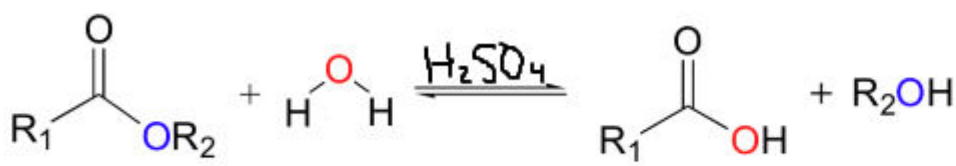
2.5 Hydrolysis Reactions

2.5.1

Hydrolysis Reactions

- These are reactions where **molecules are broken apart by adding water** (the OH part of water is added to one side of a bond and the H of water is added to the other side of a bond)
- These are the **reverse of condensation reactions** (note the reversible arrows in the reaction below)

Example of a Hydrolysis Reaction (water is being added to an ester to break it into a carboxylic acid and an alcohol):



2.6 Oxidation Reactions

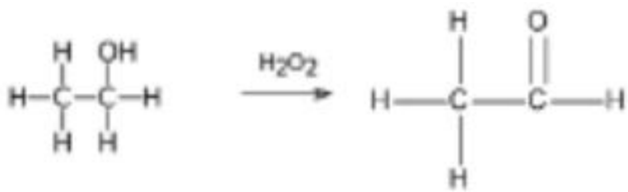
2.6.1

Oxidation vs Reduction:

Oxidation

- More/less bonds to O: ________
- More/less bonds to H: ________

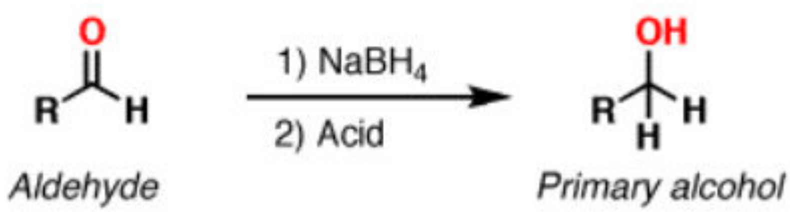
Example:



Reduction

- More/less bonds to O: ______
- More/less bonds to H: ______

Example:



Oxidizing Agents

- For oxidation reactions, we need to add **oxidizing agents that help to oxidize** the reactants to form products that have more bonds to O

- **Here are some examples of oxidizing agents:**
  - H_2O_2 (hydrogen peroxide)
  - $K_2Cr_2O_7$ (potassium dichromate)
  - $KMnO_4$ (potassium permanganate)
  - You may also see (O) written which represents an oxidizing agent

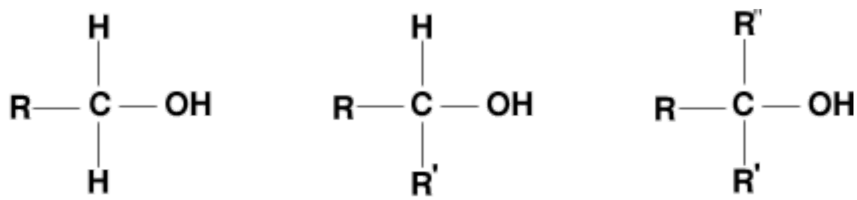
What do all of these oxidizing agents have in common?

WIZE TIP

All the oxidizing agents have a lot of O's in them. This is an easy way to remember that they are increasing the bonds to O!

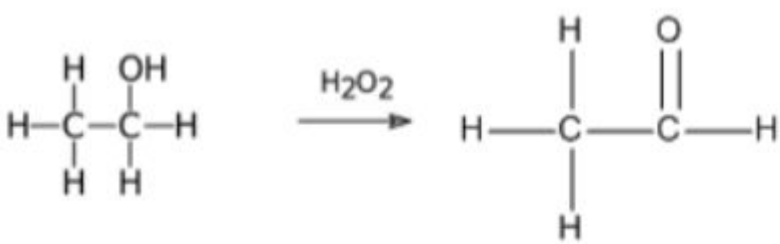
2.6.3

Oxidizing Alcohols



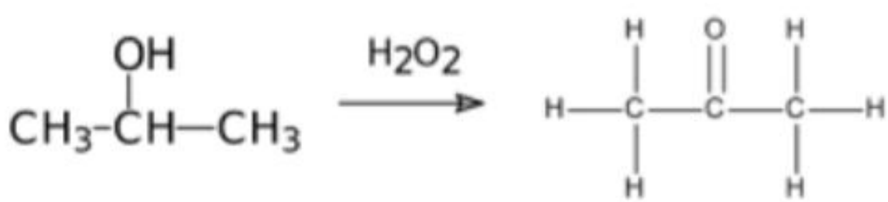
primary alcohol   secondary alcohol   tertiary alcohol

1) Primary Alcohol + Oxidizing Agent -> ____________ and Water

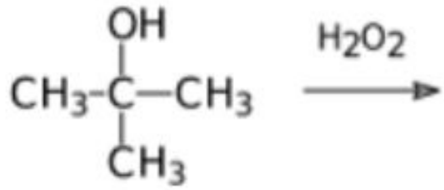


> **WIZE CONCEPT**
>
> When oxidizing alcohols, 2 Hs are removed (one from the OH and one from the adjacent C atom).
>
> We end up with a C=O group and a water molecule is produced as well!

2) Secondary Alcohol + Oxidizing Agent -> __________ and Water



3) Tertiary Alcohol + Oxidizing Agent -> __________

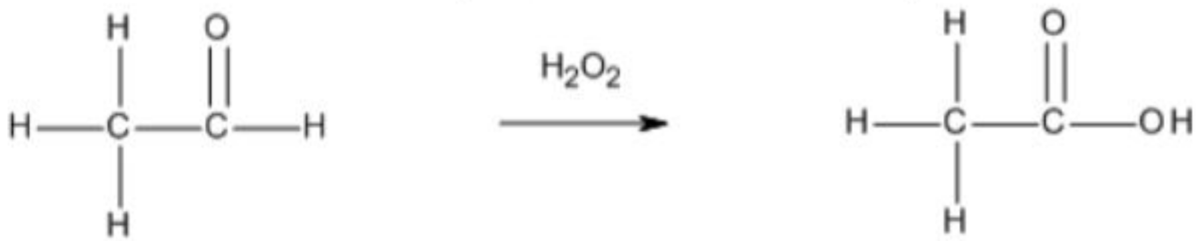

OH
CH3-C-CH3
CH3
H2O2

WATCH OUT!

Tertiary alcohols do not have an H atom available to be removed on the central carbon!
Therefore, they cannot be oxidized!
If we formed C=O, the C would have a total of 5 bonds and that doesn't make sense!

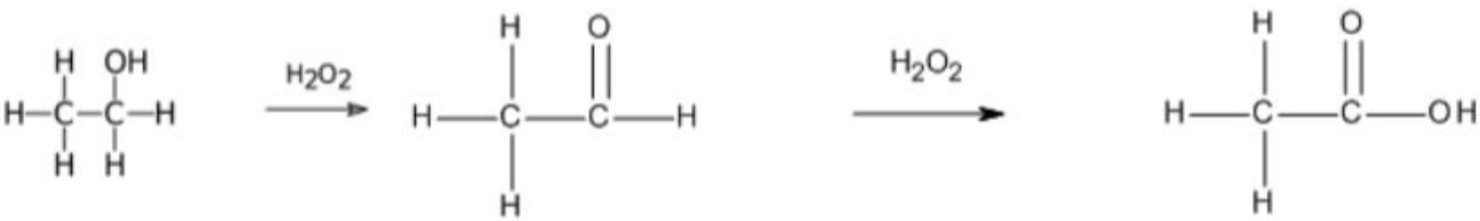
Forming Carboxylic Acids in An Oxidation Reaction

1) Reacting an Aldehyde + Oxidizing Agent -> ____________

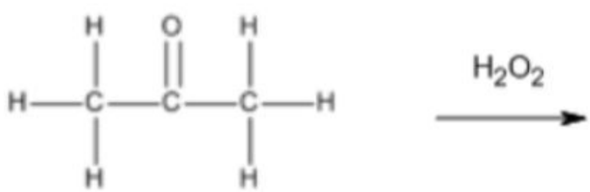


- **Review: What type of alcohol (primary, secondary, or tertiary) could be used to produce the aldehyde above?** ________

- This is showing that we can **mildly oxidize a primary alcohol to an aldehyde and further oxidize it to a carboxylic acid!**



2) Reacting a Ketone + Oxidizing Agent -> ___________



WATCH OUT!

A ketone cannot be oxidizied further!
There is no where to add a bond to O on the central C!

2.6.5

For the following reaction, draw out the reactant and product(s).

propanal + (O) -->

Provide the name of the product in the answer box:

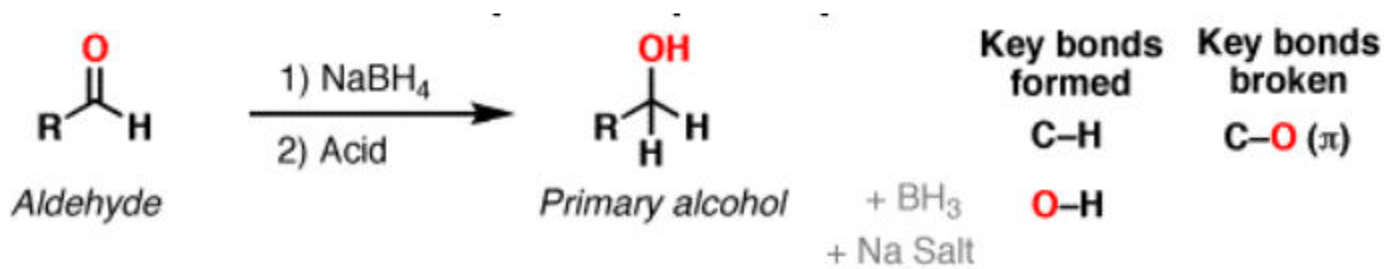
2.7 Reduction Reactions

2.7.1

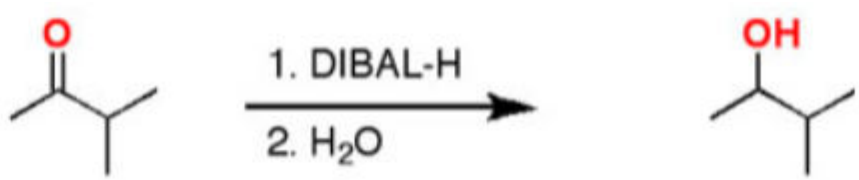
Reduction Reactions

- Note: for these reactions, high temperatures and pressures are needed, along with a catalyst
- **Review:** reduction means me will see (more/less) ________ bonds to O and (more/less) ________ bonds to H
- A common reduction agent you may seem on tests is H_2! (Adds bonds to H)

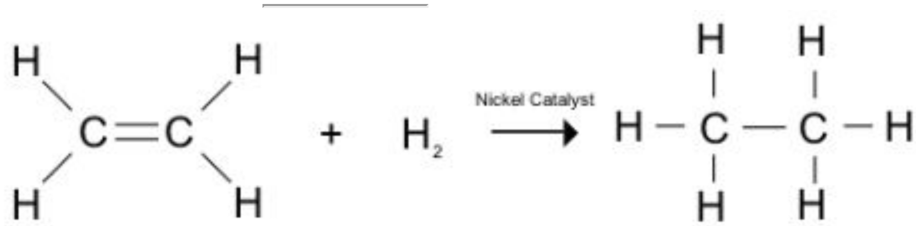
1) Reduction of an Aldehyde to form a ____________



2) Reduction of a Ketone to form a ____________



3) Review: Reduction of an Alkene -> ________ (Addition reaction aka Hydrogenation reaction)



2.8 Combustion Reactions

2.8.1

Combustion Reactions

- **Combustion reaction:** a reaction where a compound reacts with oxygen to produce the oxides of the elements that make up the compound

Complete vs Incomplete Combustion:

- **Complete Combustion:** when an **excess of oxygen** reacts with a hydrocarbon to produce CO_2 + H_2O. This reaction releases energy.

Hydrocarbon + O_2 -> CO_2 + H_2O + energy

  - Ex. Balance the following equation: _____$C_3H_8(g)$ + _____$O_2(g)$ --> _____$CO_2(g)$ + _____$H_2O(g)$

- **Incomplete Combustion:** when there is **insufficient oxygen** present and the oxygen reacts with a hydrocarbon to form CO2 + CO + H2O + C. Here, the elements in the fuel don't combine with oxygen to the greatest extent possible.

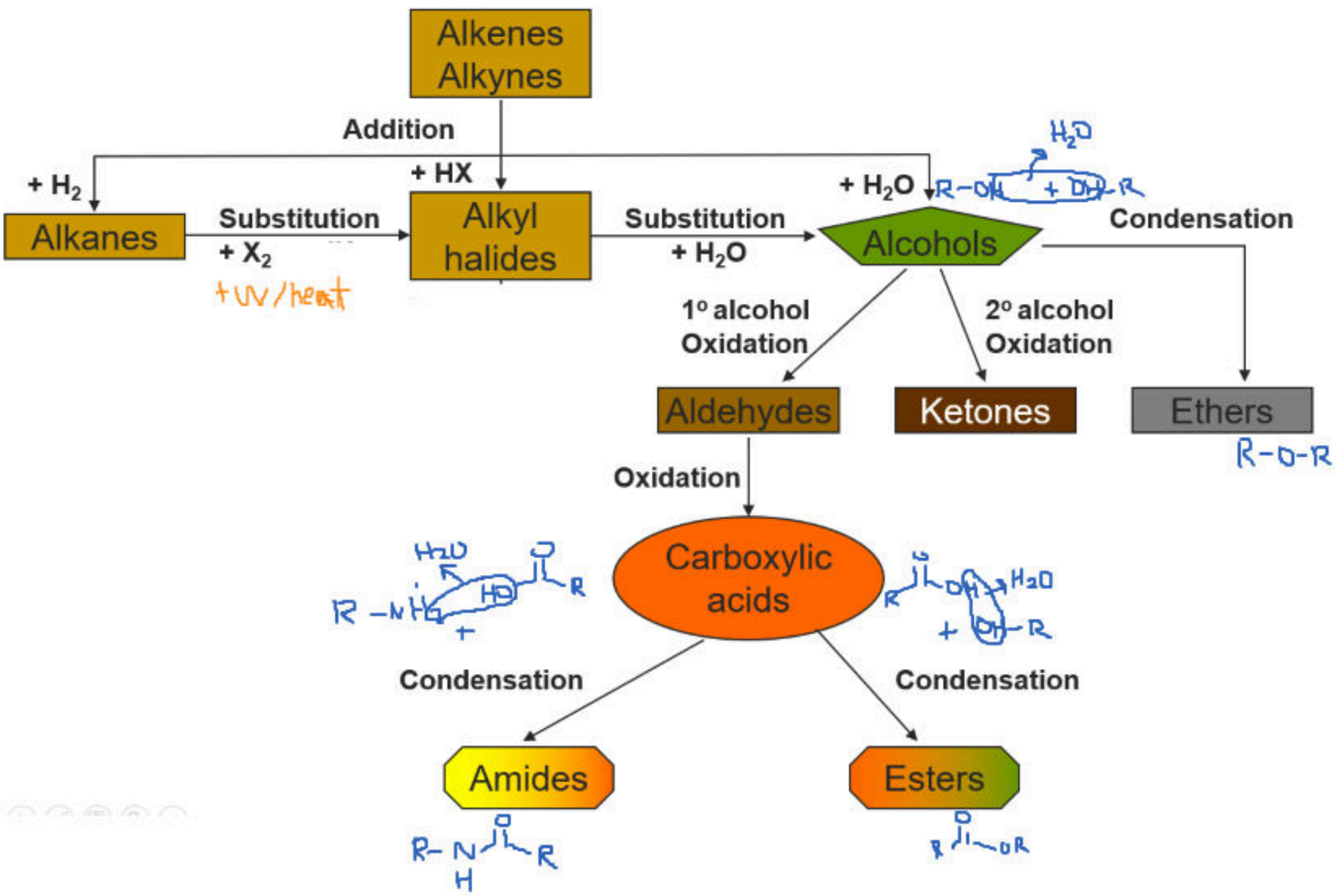
$$\text{Hydrocarbon} + O_{2(g)} \rightarrow C_{(s)} + CO_{(g)} + CO_{2(g)} + H_2O_{(g)} + \text{energy}$$

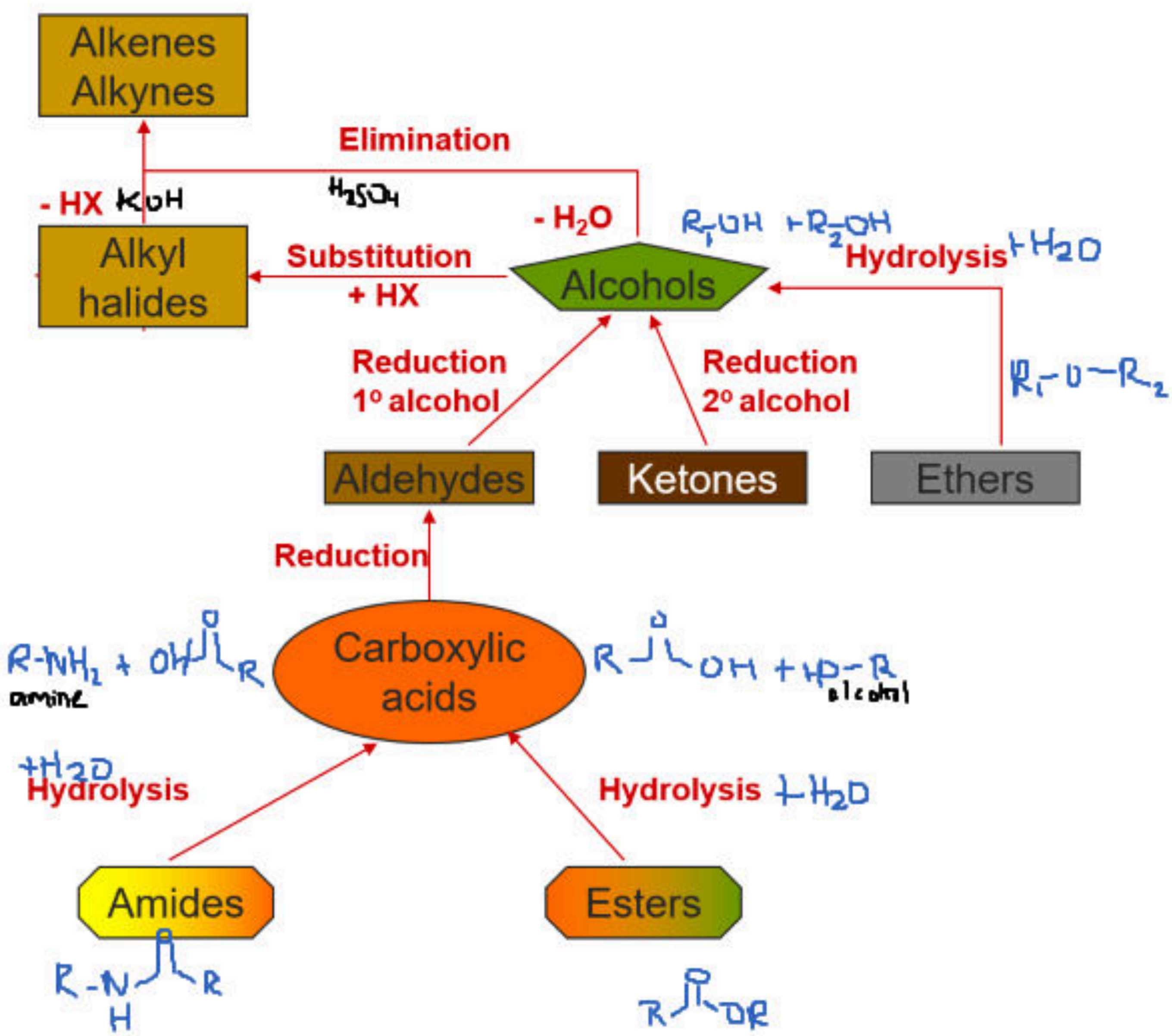
- Note: you can't balance incomplete combustion reactions :)

2.9 Summary of Reactions

2.9.1

Summary of Reactions



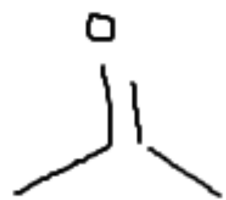


Alkenes
Alkynes
Elimination
H_2SO_4
- HX
KOH
- H_2O
Alkyl halides
Substitution
+ HX
Alcohols
$R_1OH + R_2OH$
Hydrolysis $+ H_2O$
Reduction
1° alcohol
Reduction
2° alcohol
R_1-O-R_2
Aldehydes
Ketones
Ethers
Reduction
Carboxylic acids
$R-NH_2$ + OH
amine
alcohol
$+H_2O$
Hydrolysis
Hydrolysis $+ H_2O$
Amides
Esters

2.9.2

Example: Show the series of getting from a reactant to a product

How would we go from an alkyl halide to a ketone?



2.9.3

Show the series of reactions needed to create hexan-2-one from an alkene

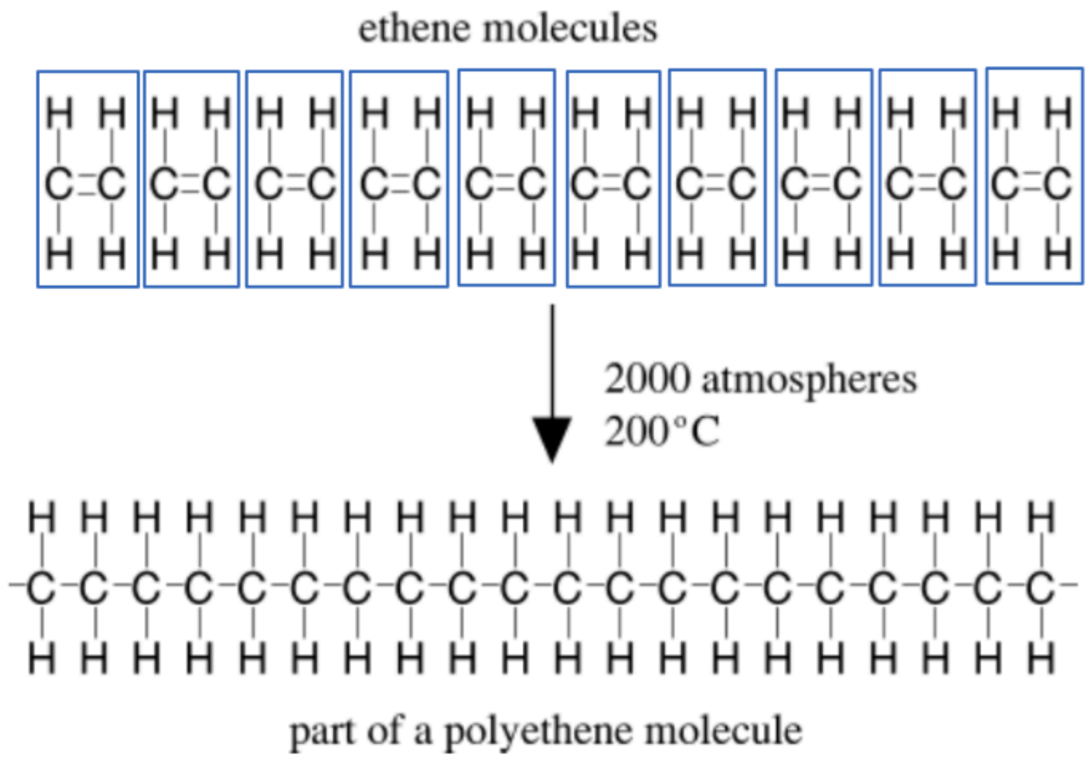
After drawing out the series, **name the alkene we started with in the answer box below:**

2.10 Polymers

2.10.1

Introduction to Polymers

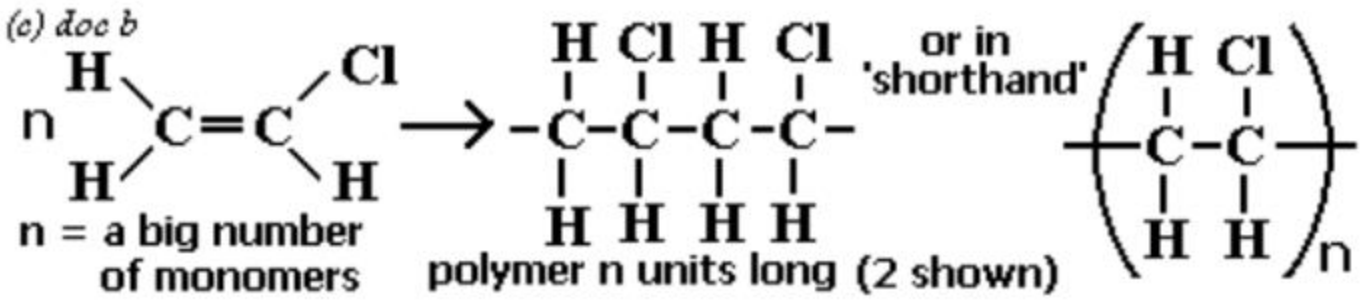
- **Polymers** are long chains composed of repeating patterns of subunits called **monomers**
- The chemical process where monomer units are joined together to form polymers is called **polymerization**



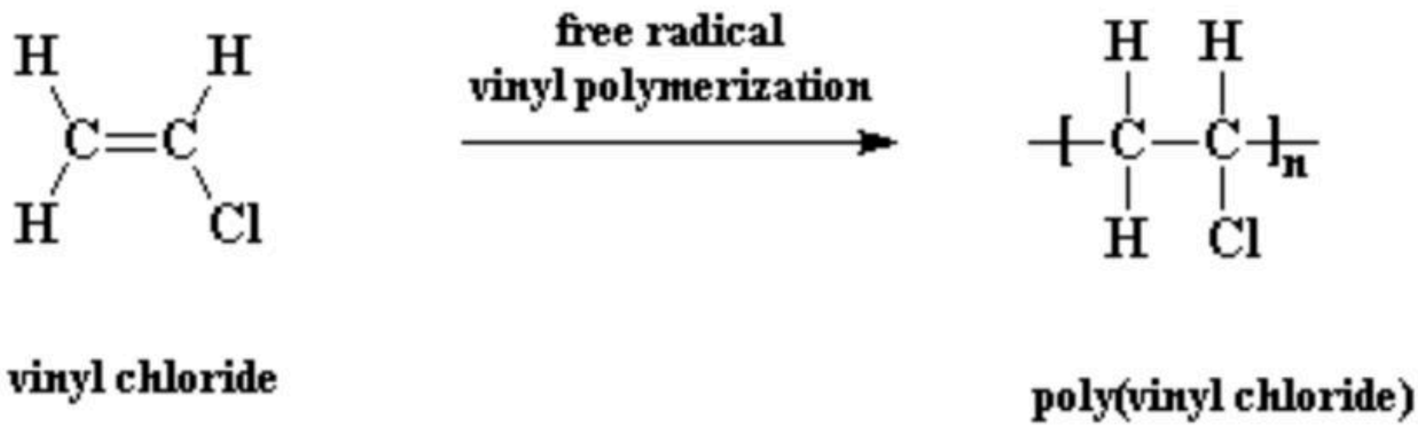
Addition Polymerization

- This is when we have **alkene monomers** that are joined together through multiple **addition reactions** to form a polymer
- Note that the polymer does not have double bonds!

Example 1:



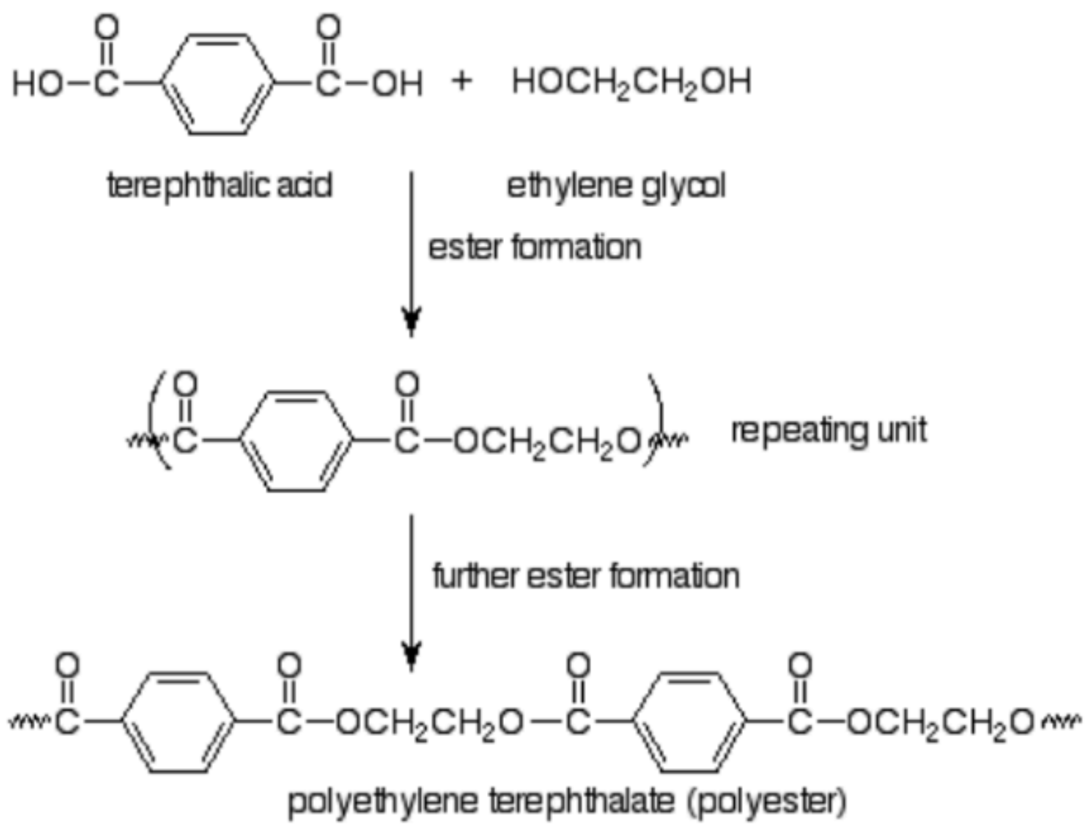
Example 2:


free radical
vinyl polymerization
H
H
C
C
H
Cl
n
vinyl chloride
poly(vinyl chloride)

Condensation Polymerization

- A condensation polymer is a polymer that is formed when **monomer units are linked by condensation reactions**
- Recall that condensation reactions result in **2 smaller molecules getting joined together and usually produces water**

Example 1:



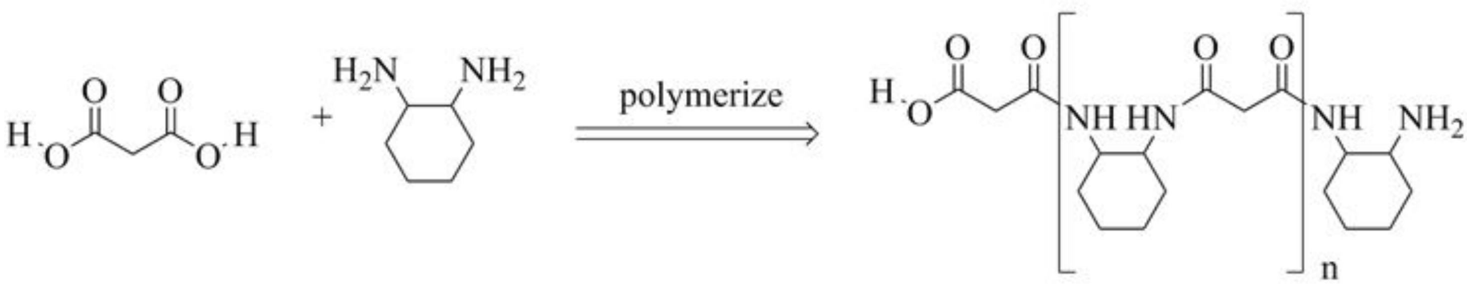
WIZE TIP

Notice that the monomer units each need to have 2 functional groups (one on each end) so that they can form a polymer and add more monomer units to each end!

In the example above, when both monomers reacted together, which functional group did we get?

For this reason we call this polymer a ____________

Example 2:



In the example above, when both monomers reacted together, which functional group did we get?

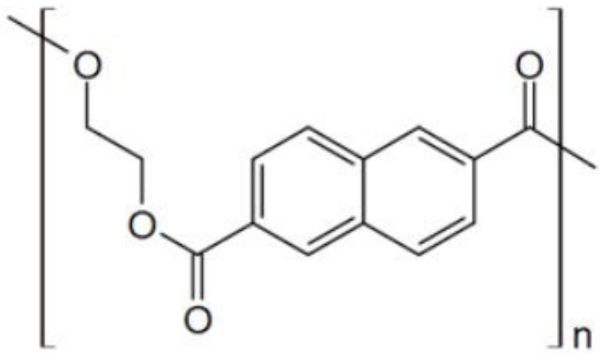
____

- For this reason, we call this polymer a ____________

2.10.4

Practice: Polymerization

Was the following polymer likely formed by addition or condensation polymerization? Draw the monomer unit(s) and any eliminated small molecule if applicable.



- addition polymerization; made of an alcohol and ester combining
- addition polymerization; made of an alcohol and a carboxylic acid combining
- condensation polymerization; made of an ether and a carboxylic acid combining
- condensation polymerization; made of an alcohol and carboxylic acid combining

2.11 Summary Checklist

2.11.1

Organic Chemistry (Part 2: Chemical Reactions) Summary Checklist

After completing this chapter you should be very familiar with the following concepts!

- Writing equations/predicting products for all types of organic reactions covered
- Predicting/ranking properties of organic compounds (eg. melting point, boiling point, solubility in various solvents)
- Polymer structures